

# CS215 Assignment-1

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## 1 Solution Q1

Let  $w_{P,k}$  represent the event that player  $P$  gets a prime number on  $k^{\text{th}}$  throw, and hence gets a win.

$$w_{P,k} = P(2, 3, 5) = \frac{3}{6} = \frac{1}{2} \text{ for both players A and B}$$

Let  $X_r$  and  $Y_r$  denote the random variables representing no of wins by  $A$  and  $B$  respectively in first  $r$  rolls. Now,

$$P(X_{n+1} > Y_n) = P(X_n > Y_n) \cdot (1) + P(X_n = Y_n) \cdot P(w_{A,n+1})$$

as either  $A$  has more wins in first  $n$  rolls itself, or  $A$  gets the last win in  $(n+1)^{\text{th}}$  roll.

Now,

$$\begin{aligned} P(X_n = Y_n = r) &= P(X_n = r) \cdot P(Y_n = r) = P(X_n = r)^2 \\ &= \left( \binom{n}{r} w^r (1-w)^{n-r} \right)^2 = \left( \binom{n}{r} \left( \frac{1}{2} \right)^r \left( \frac{1}{2} \right)^{n-r} \right)^2 \\ &= \frac{1}{2^{2n}} \cdot \binom{n}{r}^2 \end{aligned}$$

$$\begin{aligned} P(X_n = Y_n) &= \sum_{r=0}^n \frac{1}{2^{2n}} \binom{n}{r}^2 \\ &= \frac{1}{2^{2n}} \cdot \binom{2n}{n} \end{aligned}$$

By symmetry,  $P(X_n > Y_n) = P(X_n < Y_n)$

$$1 = P(X_n > Y_n) + P(X_n = Y_n) + P(X_n < Y_n)$$

$$\Rightarrow P(X_n > Y_n) = \frac{1 - P(X_n = Y_n)}{2}$$

$$\Rightarrow P(X_n > Y_n) = \frac{1}{2^{2n+1}} \cdot \left( 2^{2n} - \binom{2n}{n} \right)$$

So, finally

$$\begin{aligned} P(X_{n+1} > Y_n) &= \frac{1}{2^{2n+1}} \cdot \left( 2^{2n} - \binom{2n}{n} \right) + \frac{1}{2^{2n}} \cdot \binom{2n}{n} \cdot \frac{1}{2} \\ &= \frac{1}{2} \end{aligned}$$

## 2 Solution 4

(a)  $E_i$  is the event that  $i^{th}$  candidate is the best and we hire him. For this to happen, two conditions have to be met:

1.  $i^{th}$  candidate is the best among all the  $n$  appeared, the probability of which is given by:

$$P(i^{th} \text{ is the best}) = \frac{1}{n} \text{ (All are equally likely to be the best)}$$

2.  $i^{th}$  candidate is better than all of the previously candidates being interviewed and the probability of hiring this candidate is given by:

$$P(\text{hired} \mid i^{th} \text{ is better than previous candidates}) = \frac{m}{i-1}$$

as this candidate must not be among first  $m$  rejected out of previously  $i-1$  interviewed.

Thus by the definition of independent probability,

$$P_r(E_i) = P(i^{th} \text{ is the best and } i^{th} \text{ is hired}) = \frac{m}{n(i-1)} \quad (1)$$

Now, to determine the total probability that we hired the best candidate, sum over  $P_r(E_i)$  from  $i = 0$  to  $i = n$ :

$$P_r(E) = \sum_{i=0}^n P_r(E_i)$$

$$\Rightarrow P_r(E) = \sum_{i=0}^m P_r(E_i) + \sum_{i=m+1}^n P_r(E_i)$$

$$\Rightarrow P_r(E) = 0 + \frac{m}{n} \sum_{i=m+1}^n \frac{1}{i-1} \text{ (Starting } m \text{ candidates were rejected so there probability is 0)}$$

$$\text{Hence, } P_r(E) = \frac{m}{n} \sum_{i=m+1}^n \frac{1}{i-1}$$

(b) We can establish the summation bound using integral generalisation as follows:

$$\sum_{j=m+1}^n \frac{1}{j} \leq \sum_{j=m+1}^n \frac{1}{j-1} \leq \sum_{j=m}^n \frac{1}{j-1}$$

$$\Rightarrow \frac{m}{n} \int_m^n \frac{1}{x} dx \leq P_r(E) \leq \frac{m}{n} \int_m^n \frac{1}{x-1} dx$$

$$\Rightarrow \frac{m}{n} [ln|x|]_m^n \leq P_r(E) \leq \frac{m}{n} [ln|x-1|]_m^n$$

Therefore, we have the bounds  $\frac{m}{n} (ln(n) - ln(m)) \leq P_r(E) \leq \frac{m}{n} (ln(n-1) - ln(m-1))$

(c) Let

$$f(m) = \frac{m}{n} (ln(n) - ln(m))$$

The above function will be maximised when  $f'(m) = 0$  and  $f''(m) < 0$ .

$$f'(m) = \frac{ln(n)}{n} - \frac{1}{n} [ln(m) + 1]$$

Solving the above expression for 0, we get,

$$m = \frac{n}{e}$$

Thus the function is maximized for above value of m and,

$$P_r(E) \geq \frac{1}{e} (ln(e))$$

Hence,  $P_r(E) \geq \frac{1}{e}$

### 3 Solution Q5

Assume that the IDs of the traders are randomly distributed.

Let the probability of me winning if I choose my position to be  $r$  be  $F(r)$ .

$$F(r) = P(\text{I win} \mid \text{I am at position } r) = P(\text{Traders at } 1, 2, \dots, r-1 \text{ lose}) \times P(\text{I win at position } r)$$

$$F(r) = \begin{cases} 0 & \text{if } r = 1, \\ \frac{200}{200} \cdot \frac{199}{200} \cdot \dots \cdot \frac{200-(r-2)}{200} \cdot \frac{r-1}{200} & \text{if } 2 \leq r \leq 200, \\ 0 & \text{if } r > 200. \end{cases}$$

Now, suppose my winning probability is maximum at  $r = r_0$ .

Then,  $F(r) > F(r-1)$  and  $F(r) > F(r+1)$ .

$$\begin{aligned} \frac{200}{200} \dots \frac{203-r}{200} \cdot \frac{202-r}{200} \cdot \frac{r-1}{200} &> \frac{200}{200} \dots \frac{203-r}{200} \cdot \frac{r-2}{200} \\ \frac{200}{200} \dots \frac{202-r}{200} \cdot \frac{r-1}{200} &> \frac{200}{200} \dots \frac{202-r}{200} \cdot \frac{201-r}{200} \cdot \frac{r}{200} \end{aligned}$$

$$\begin{aligned} \frac{202-r}{200} \cdot (r-1) &> (r-2) & \frac{201-r}{200} \cdot r &< r-1 \\ \Rightarrow r^2 - 203r + 202 + 200r - 400 &< 0 & \Rightarrow r^2 - 201r + 200r - 200 &> 0 \\ \Rightarrow r^2 - 3r - 198 &< 0 & \Rightarrow r^2 - r - 200 &> 0 \\ \Rightarrow r \in (-12.65, 15.65) & & \Rightarrow r \in (-\infty, -13.65) \cup (14.65, \infty) \end{aligned}$$

Hence, by taking the intersection of the above intervals, I conclude that the probability of me winning is maximum at  $r = 15$ .

### 4 Solution 6

(a) We define mean as,

$$\begin{aligned} \text{Mean} &= \frac{\sum x_i}{n} \\ \Rightarrow \sum x_i &= \text{mean} * n \\ \Rightarrow \text{NewSum} &= \sum x_i + \text{NewDataValue} \\ \Rightarrow \text{NewMean} &= \frac{\text{NewSum}}{n+1} \end{aligned}$$

$$\Rightarrow NewMean = \frac{mean * n + NewDataValue}{n + 1}$$

This is how we can update mean using previous mean and new data value without recalculating mean again.

(b) However in case of median, there will be cases when n will be even or odd and we will be needing A[index] to access required value for median. After appending new data value to the list A and sorting it,

$$NewMedian = \begin{cases} A[n/2] & \text{if n is even,} \\ \frac{OldMedian + A[\frac{n-1}{2} + 1]}{2} & \text{if n is odd.} \end{cases}$$

(c) By the definition of standard Deviation,

$$OldStd = \sqrt{\frac{\sum (x_i - oldMean)^2}{n - 1}}$$

$$\Rightarrow (n - 1)(OldStd)^2 = \sum (x_i - oldMean)^2$$

$$\Rightarrow (n - 1)(OldStd)^2 = \sum x_i^2 - 2OldMean \sum x_i + n(oldMean)^2$$

$$\Rightarrow (n - 1)(OldStd)^2 = \sum x_i^2 - n(oldMean)^2$$

$$\Rightarrow OldSumOfSquare = (n - 1)(OldStd)^2 + n(oldMean)^2$$

$$\Rightarrow NewSumOfSquare = (n - 1)(OldStd)^2 + n(oldMean)^2 + (NewDataValue)^2$$

$$\Rightarrow NewSumOfSquare - (n+1)(NewMean)^2 = (n-1)(OldStd)^2 + n(oldMean)^2 + (NewDataValue)^2 - (n+1)(NewMean)^2$$

$$\Rightarrow NewStd = \sqrt{\frac{(n - 1)(OldStd)^2 + n(oldMean)^2 + (NewDataValue)^2 - (n + 1)(NewMean)^2}{n}}$$

This is the derivation for updated standard deviation using old standard deviation, old and New mean.

In order to update the histogram, first search the new data point bin and then increase the count of bin by 1.